CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

Identification of SAD Methodology

This project adopts the Structured Systems Analysis and Design Method (SSADM) as the approach for analyzing and designing the system. SSADM is suitable for this system because it focuses on structured processes, data flow, and logical organization of system components. The system being developed is data-driven and involves clearly defined operations such as product management, customer handling, supplier management, and sales processing. SSADM provides a systematic way of understanding how data moves within the system and how different processes interact with the database. It also allows separation between logical design and physical implementation, making it easier to develop a well-organized and maintainable system.

Analysis of Collected Data Using Selected Method

The analysis of the collected data shows that the system is designed to manage inventory and sales operations through several key data entities. These include users, products, categories, customers, suppliers, and sales. Users interact with the system and perform operations such as adding products, recording sales, and managing customer and supplier information. The products entity is central to the system and contains details such as product name, generic name, category, supplier, original price, selling price, quantity, as well as arrival and expiry dates. The inclusion of arrival and expiry dates enables the system to track product lifecycle and monitor stock validity.

Categories are used to group products for easier organization, while customers and suppliers store information about individuals or organizations involved in transactions. The sales entity captures transaction details including the product sold, customer, quantity, payment, and date of sale. Data flows from the user into the system through input forms, is processed by the

system, and is then stored in the respective database tables. The system also retrieves stored data to display information such as product listings, sales records, and reports.

System Requirement Specification

The system requirements define what the system is expected to do and how it should perform. The functional requirements of the system include the ability to allow users to log in and log out, manage user accounts, and perform operations such as adding, updating, deleting, and viewing products, categories, customers, and suppliers. The system should also record sales transactions, calculate profit or loss based on product prices, update product quantity after each sale, and generate reports. In addition, the system must record product arrival and expiry dates and support monitoring of expired or near-expiry products.

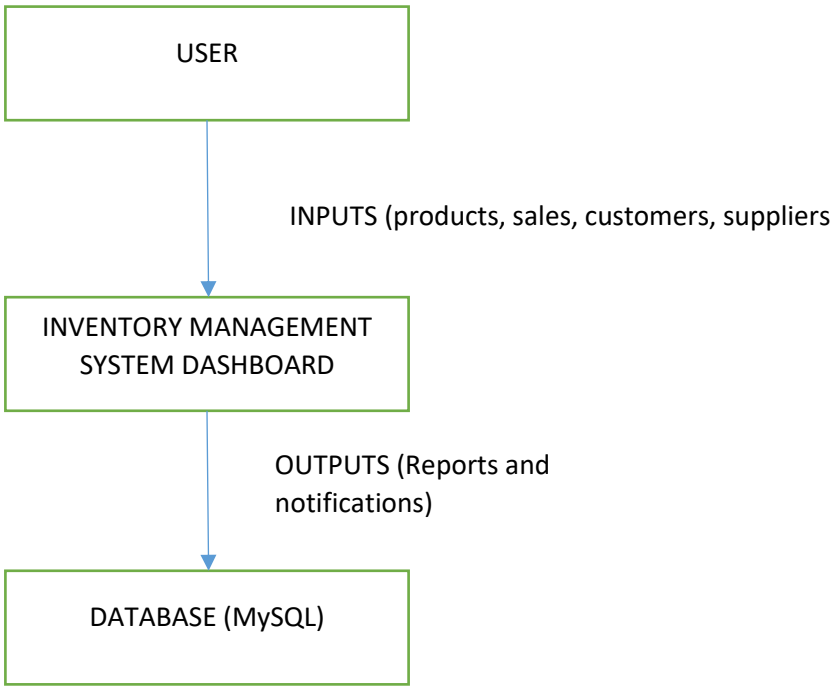
The non-functional requirements focus on system quality attributes. The system should ensure security by requiring user authentication and protecting user data. It should provide good performance by responding quickly to user actions and processing data efficiently. Usability is also important, and the system should have a simple and user-friendly interface. Reliability is required to ensure that data is stored correctly and errors are handled appropriately. The system should also be maintainable so that it can be updated and improved easily when needed.

Design of the System According to Requirements Specified

Logical Design

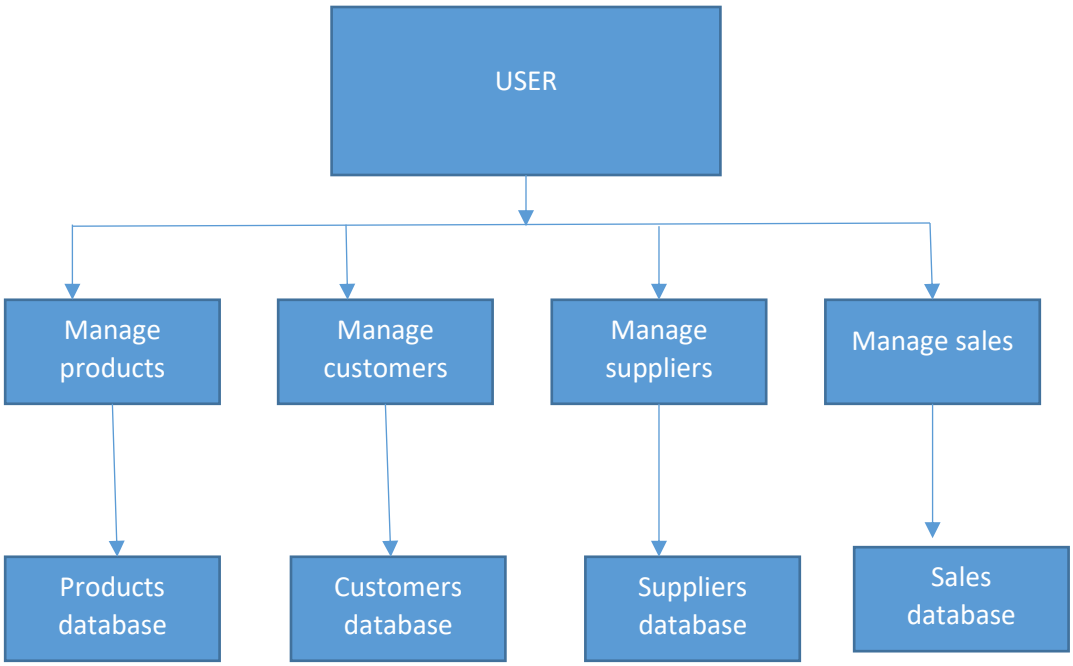
The logical design represents how data flows within the system without focusing on physical implementation. It shows how users interact with the system and how data moves between processes and data stores.

Context Diagram (Level 0 DFD)



The context diagram shows the system as a single process interacting with the external entity, which is the user. The user provides input such as product details, customer information, and sales data. The system processes this input and stores it in the database, and then provides output such as reports and notifications back to the user.

Level 1 Data Flow Diagram (DFD)



The Level 1 DFD breaks the system into major processes. These include managing products, customers, suppliers, and sales. Each process receives input from the user, processes the data, and stores it in the respective database table. The product management process includes handling important fields such as arrival date and expiry date, which are essential for inventory tracking.

Physical Design

The physical design describes how the system is implemented in terms of inputs, outputs, storage, processing, and control.

Input Requirements: The system accepts input through various forms. These include product forms, customer forms, supplier forms, and sales forms. The product form includes fields such as product name, generic name, category, supplier, arrival date, expiry date, original price, selling price, and quantity. These inputs ensure that all necessary product details are captured for proper inventory management.

Output Requirements: The system generates outputs such as product lists, sales reports, customer records, and inventory summaries. It also provides notifications to inform users about successful or failed operations. Additionally, the system supports exporting data to formats such as Excel for reporting purposes.

Storage Requirements: The system uses a relational database to store data. The database consists of tables for users, products, categories, customers, suppliers, and sales. Each table stores structured data relevant to system operations. The product table includes fields for arrival and expiry dates, which help in tracking the product lifecycle. Although these fields are currently stored as text, it is recommended that they be stored using proper date data types for improved data integrity and validation.

Processing Requirements: The system performs several processing activities including validation of user input, calculation of profit or loss, updating product quantities after sales, and handling communication between the user interface and the server. It also processes date-related information such as arrival and expiry dates to ensure accurate inventory tracking.

System Control and Backup: The system includes control mechanisms such as user authentication, input validation, and notification handling to ensure proper system operation. Backup and recovery can be achieved through regular database exports, which help prevent data loss and ensure system reliability.

Detailed Design

User Interface Design

The user interface is designed to be simple and user-friendly, allowing users to interact with the system easily. It includes dashboards and forms for managing products, customers, suppliers, and sales. The product interface allows users to input important details such as arrival

and expiry dates, ensuring complete tracking of product information. After login into the system, dashboard is displayed first, showing numerous things that point to the functionalities of the system. For instance, it shows a column in the left that has the menu which lists important features like creating product, which is basically adding a product into the system such that they are managed through the system instead of through the manual system. The menu also lists reports which presents all the reports about sales, stock value, customers, top selling products, payments, stock alerts, payment methods etc. The menu contains the users, including the admin and the recognized users who are permitted to use the system. This section has the suppliers who supply drugs to Elkim, customers, and the system users who are basically the employees. The settings are also in the dashboard, where system settings, backup, system update, and restart are. Then there is a logout form, which permits logging out of the system. The dashboard is user-friendly, such that when users interact with the system, they just need to click the function they need to perform and fill in the details. For instance, drug categories such as antibiotics, antifungals etc are created only through clicking “add category” and thereafter, they can add products into such categories easily as they need to click “add products” and a pop-up window-like form appears on the screen with fields open for filling the details of a product and after filling these details, they click save and product is added into the system and classified into its category.

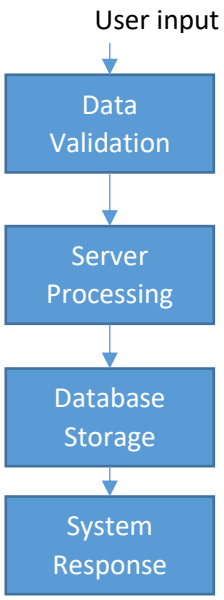
Data Design

The data design describes how data is structured within the system. The system uses multiple tables, each representing a specific entity such as users, products, categories, customers, suppliers, and sales. Each table includes a primary key for identification and contains relevant attributes. The design recommends the use of appropriate data types, such as DATE, for storing

arrival and expiry dates, as well as the use of relationships between tables to improve consistency and reduce redundancy.

Process Design

The process design explains how data flows through the system from input to storage and output.



The process begins when the user enters data into the system through input forms. The system validates the data to ensure correctness before sending it to the server. The server processes the data and stores it in the database. Once the operation is complete, the system sends a response back to the user in the form of a notification. This ensures that all data is properly handled and that users receive feedback on their actions. For instance, when a user clicks “add product” it opens a field where the details of the product such as product name, category, generic name, the supplier, quantity and date it is added and the date it expires are filled. The user clicks save and if one field is not filled, the system returns a notification to fill it. If the data is correct, it is validated and server processes it and stores it in the database and a notification is send “success! Operation complete.” This means that the product has been added into the system successfully and its components are stored within the database.